

Chapter 2 - Sheaves

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1 Motivating example: The sheaf of smooth functions

Exercise 1.A

Show that this is the only maximal ideal of O_p . (Hint: show that every element of $O_p \setminus \mathfrak{m}_p$ is invertible.)

Proof. For $\forall (f, U) \in \mathcal{O}_p \setminus \mathfrak{m}_p$, we have

$$\left(\frac{1}{f}, U'\right) \in \mathcal{O}_p \setminus \mathfrak{m}_p$$

in a neighbourhood U' of p , hence invertible. □

Exercise 1.B [Skipped. I haven't learnt Differential Geometry yet]

EXERCISE FOR THOSE WITH DIFFERENTIAL GEOMETRIC BACKGROUND.

Prove this. (Rhetorical question for experts: what goes wrong if the sheaf of continuous functions is substituted for the sheaf of smooth functions? What goes wrong if you use the sheaf of C^1 functions?)

2 Definition of sheaf and presheaf

Exercise 2.A [Trivial]

EXERCISE FOR CATEGORY-LOVERS: "A PRESHEAF IS THE SAME AS A CONTRAVARIANT FUNCTOR".

Given any topological space X , we have a "category of open sets" (Example 1.1.9), where the objects are the open sets and the morphisms are inclusions. Verify that the data of a presheaf is precisely the data of a contravariant functor from the category of open sets of X to the category of sets. (This interpretation is surprisingly useful.)